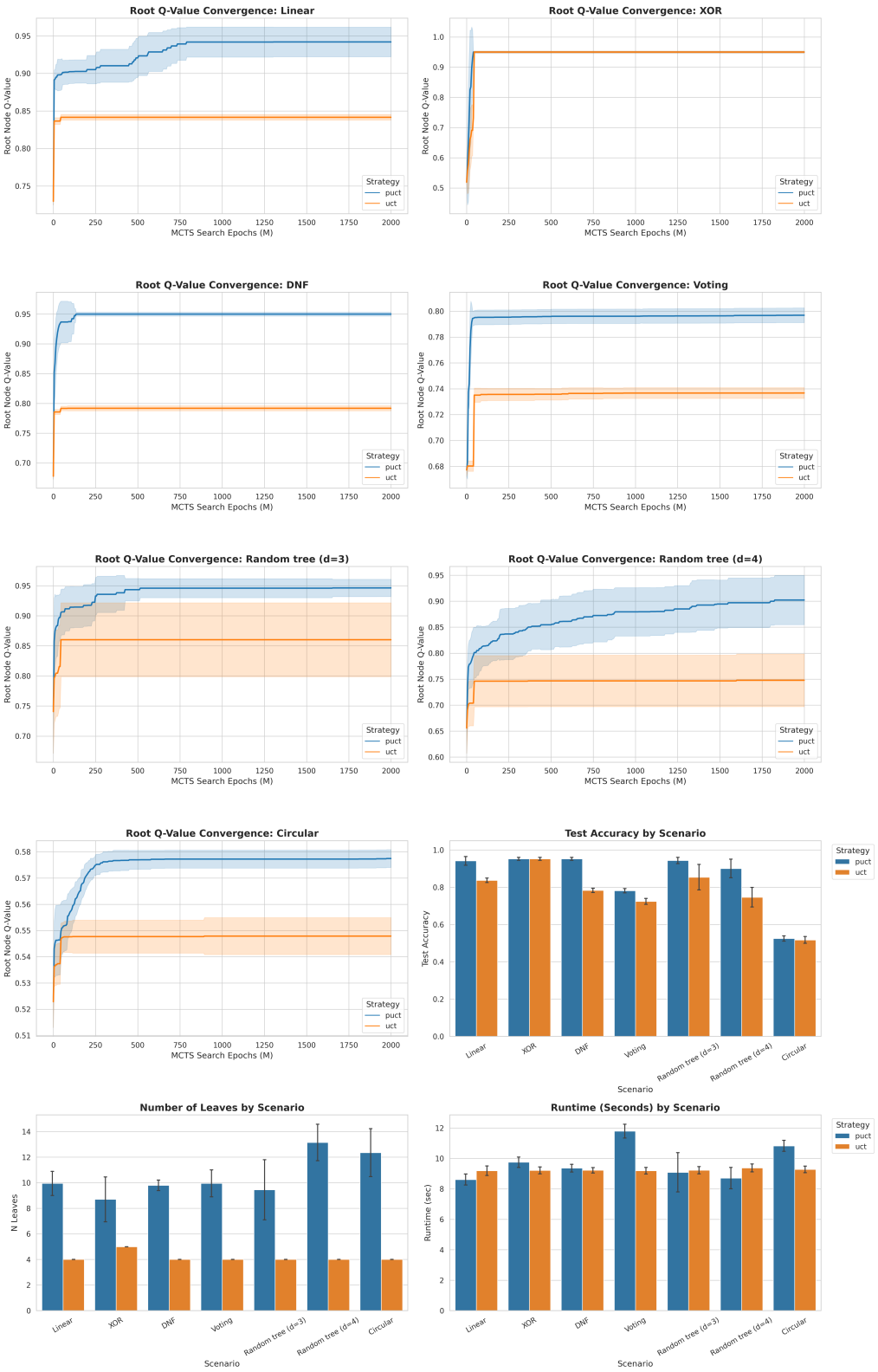

Figure 1: Comprehensive evaluation of MCTS strategies with and without prior guidance, showing Root Q-Value convergence, test accuracy, number of leaves, and runtime across different scenarios.

** Note: In the legend, "puct" denotes the strategy utilizing prior probabilities to guide exploration, while "uct" serves as an ablation baseline without prior guidance (effectively setting all prior probabilities to 1). Experimental settings are fixed at maximum depth $d = 4$, number of variables $V = 40$, sample size $N = 5000$, binary features, binary classification, and noise level 0.05. All results are averaged over 20 independent trials.*

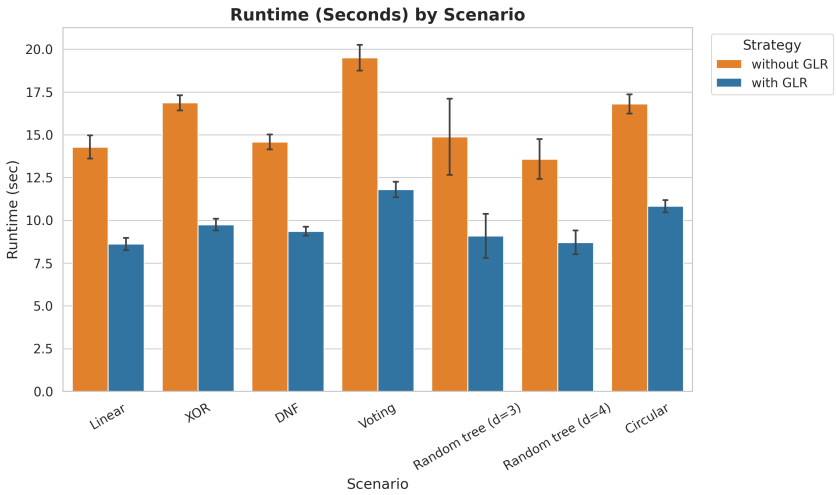

Figure 2: Ablation study demonstrating the runtime acceleration achieved by the Global Leaf Registry (GLR) across various scenarios.

** Note: Experimental settings are fixed at maximum depth $d = 4$, number of variables $V = 40$, sample size $N = 5000$, binary features, binary classification, and noise level 0.05. All results are averaged over 20 independent trials.*

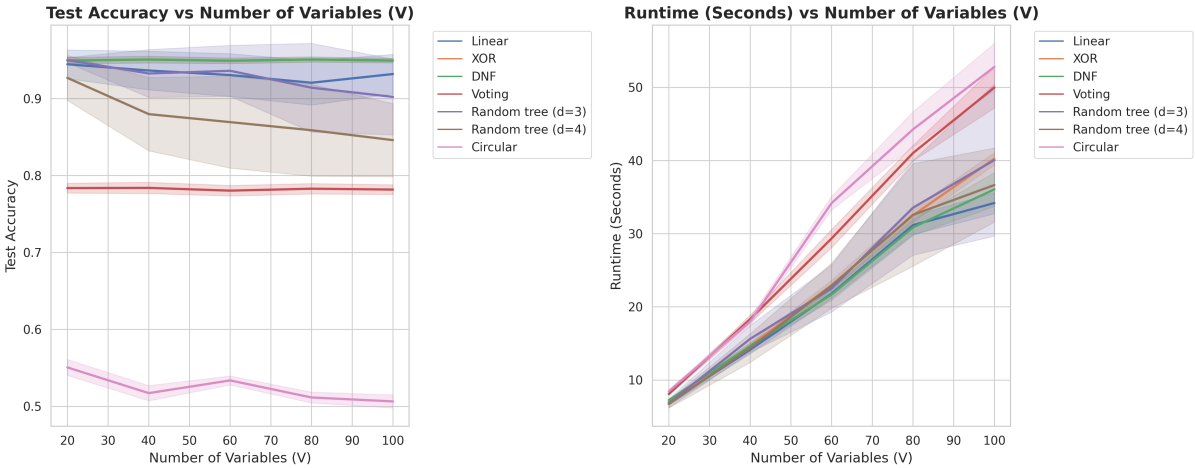

Figure 3: Scalability analysis: Test accuracy and runtime with respect to the number of variables (V).

** Note: The sample size is fixed at $N = 20000$ and maximum depth is $d = 4$. Other settings include binary features, binary classification, and noise level 0.05. Results are reported as Mean ± 95% Confidence Interval (CI) over 20 independent trials.*

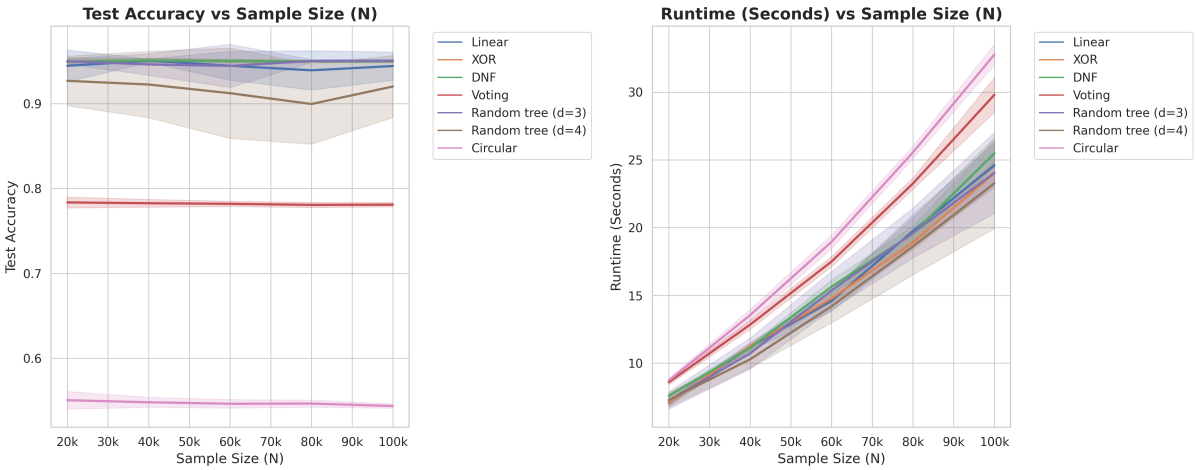

Figure 4: Scalability analysis: Test accuracy and runtime with respect to the sample size (N).

** Note: The number of variables is fixed at $V = 20$ and maximum depth is $d = 4$. Other settings include binary features, binary classification, and noise level 0.05. Results are reported as Mean ± 95% Confidence Interval (CI) over 20 independent trials.*